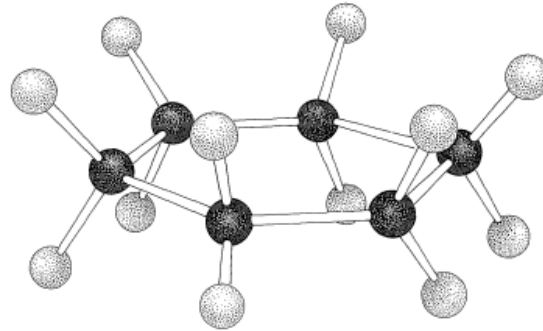


CONFORMERS OF CYCLOHEXANE

Conformations of Cyclohexane

A **planar structure** for cyclohexane is clearly improbable. The bond angles would necessarily be **120°**, **10.5° larger than the ideal tetrahedral angle**. Also, **every carbon-hydrogen bond** in such a structure **would be eclipsed**. The resulting angle and eclipsing strains would severely destabilize this structure. The ring strain of planar cyclohexane is in excess of 84 kJ/mol so it rarely discussed other than in theory.



IDEA OF CHAIR/BOAT FORM:

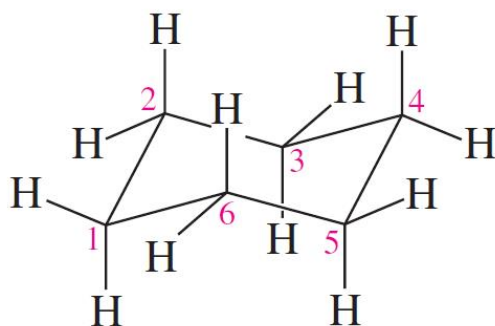
Following the recognition of the tetrahedral nature of saturated carbon in 1874 (or perhaps 1869), **Sachse in 1890** using only trigonometry, derived the **non-planar chair and boat conformations** of cyclohexane and the **axial and equatorial positions** of the hydrogens.

Sachse-Mohr theory

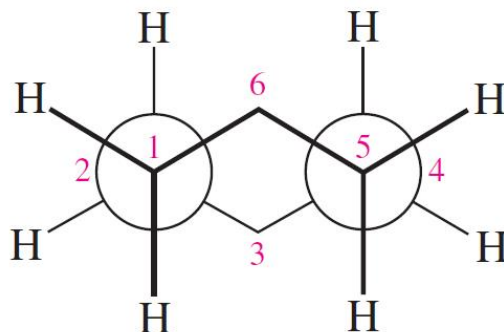
- Ernst Mohr (1873-1926) took up the hypothesis of Sachse in 1918 and applied it to Decalin. He assumed the existence of two isomers, the *cis*- and the *trans*-decalin, both of which could be isolated by the German chemist Walter Hückel (1895-1973) in 1925.
- It is now generally accepted that the Sachse-Mohr theory is correct.

CONFORMERS OF CYCLOHEXANE

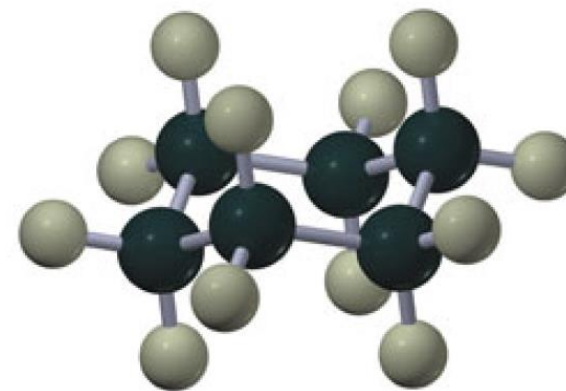
- The cyclic compounds most commonly found in nature contain six-membered rings because carbon rings of that size can exist in a conformation—called a **chair conformer**—that is almost completely free of strain.
- All the bond angles in a **chair conformer** are **111°** (which is very close to the ideal tetrahedral bond angle of **109.5°**) and all the adjacent bonds are **staggered**



chair conformer of
cyclohexane



Newman projection of
the chair conformer
looking down the
C-1—C-2 and C-5—C-4 bonds



ball-and-stick model of
the chair conformer

How to draw the chair conformer of cyclohexanes:



A



B

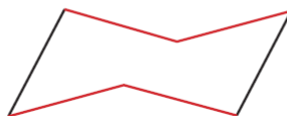


C

1. Draw two parallel lines of the same length, slanted upward.

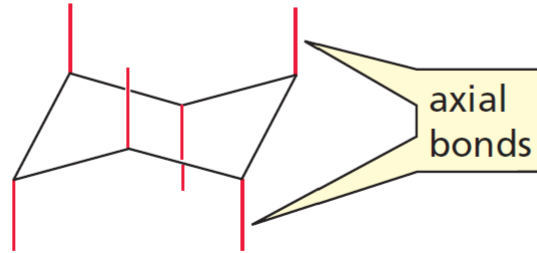


2. Connect the tops of the lines with a V whose left side is slightly longer than its right side. Connect the bottoms of the lines with an inverted V. (The bottom-left and top-right lines should be parallel; the top-left and bottom-right lines should be parallel.) This completes the framework of the six-membered ring.

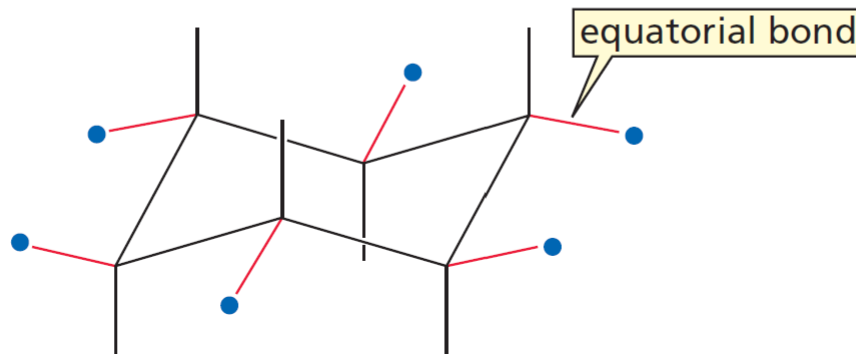


How to draw the chair conformer of cyclohexanes:

3. Each carbon has an axial bond and an equatorial bond. The **axial bonds** (red lines) are vertical and alternate above and below the ring. The axial bond on one of the uppermost carbons is up, the next is down, the next is up, and so on.

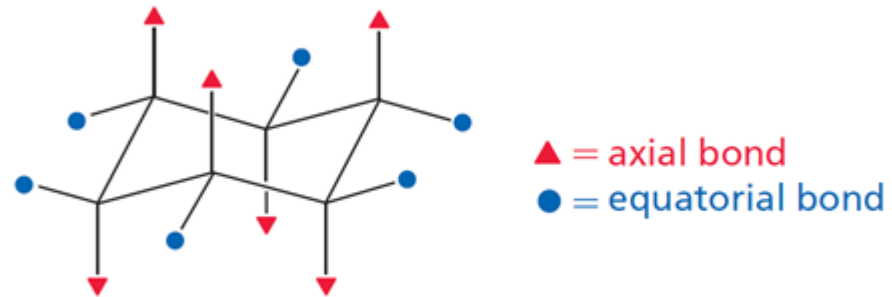


4. The **equatorial bonds** (red lines with blue balls) point outward from the ring. Because the bond angles are greater than 90° , the equatorial bonds are on a slant. If the axial bond points up, then the equatorial bond on the same carbon is on a downward slant. If the axial bond points down, then the equatorial bond on the same carbon is on an upward slant.



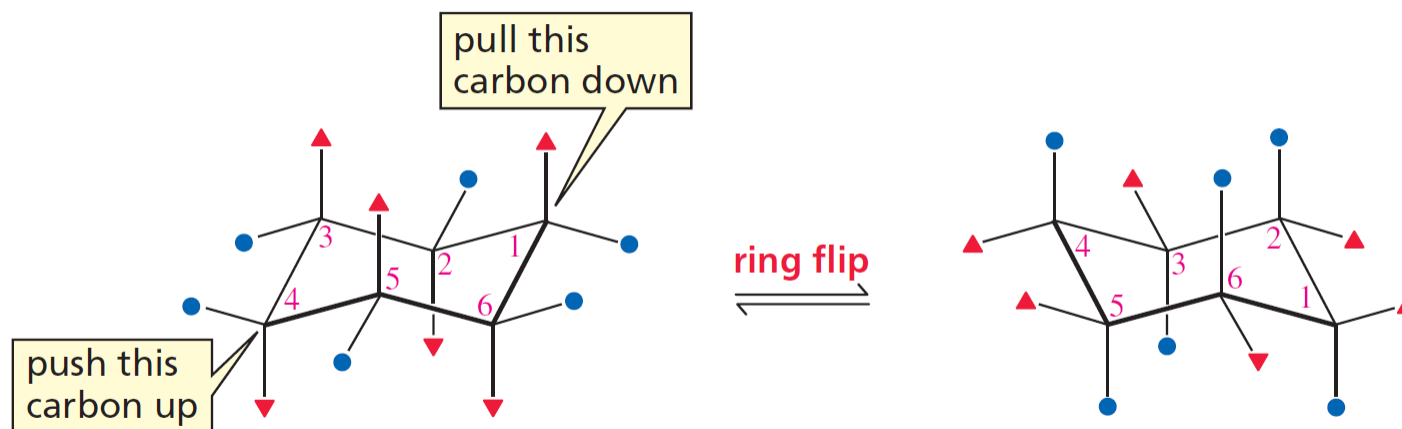
How to draw the chair conformer of cyclohexanes:

Notice that each equatorial bond is parallel to two ring bonds (red lines) one bond away.



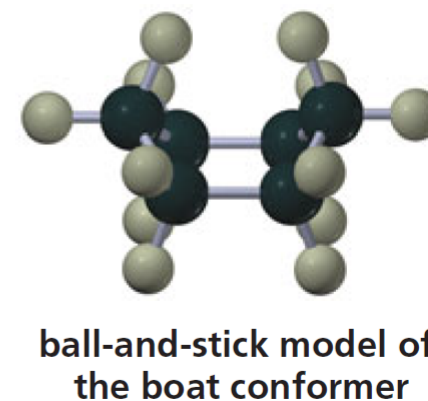
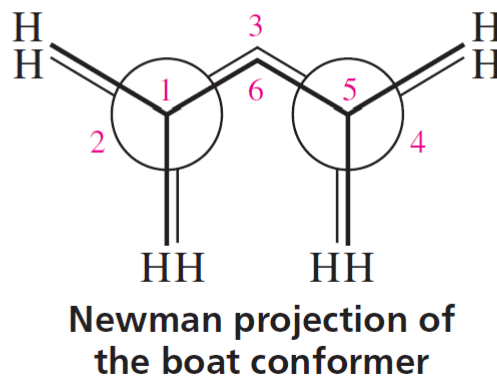
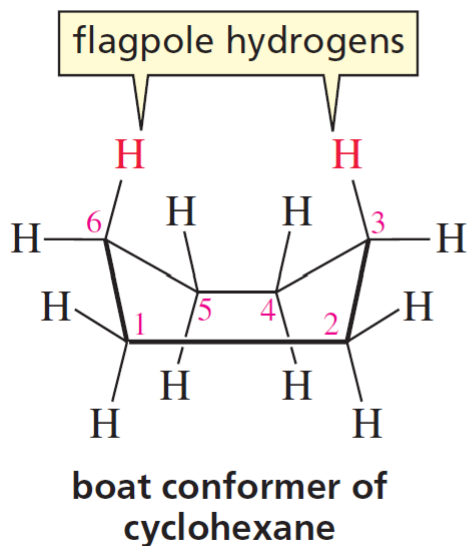
CONFORMERS OF CYCLOHEXANE

- **Cyclohexane** rapidly interconverts between **two stable chair conformers** because of the ease of rotation about its **C-C** bonds.
- This interconversion is called **ring flip**
- When the two chair conformers interconvert, bonds that are **equatorial** in one chair conformer become **axial** in the other chair conformer, and bonds that are **axial** become **equatorial**.



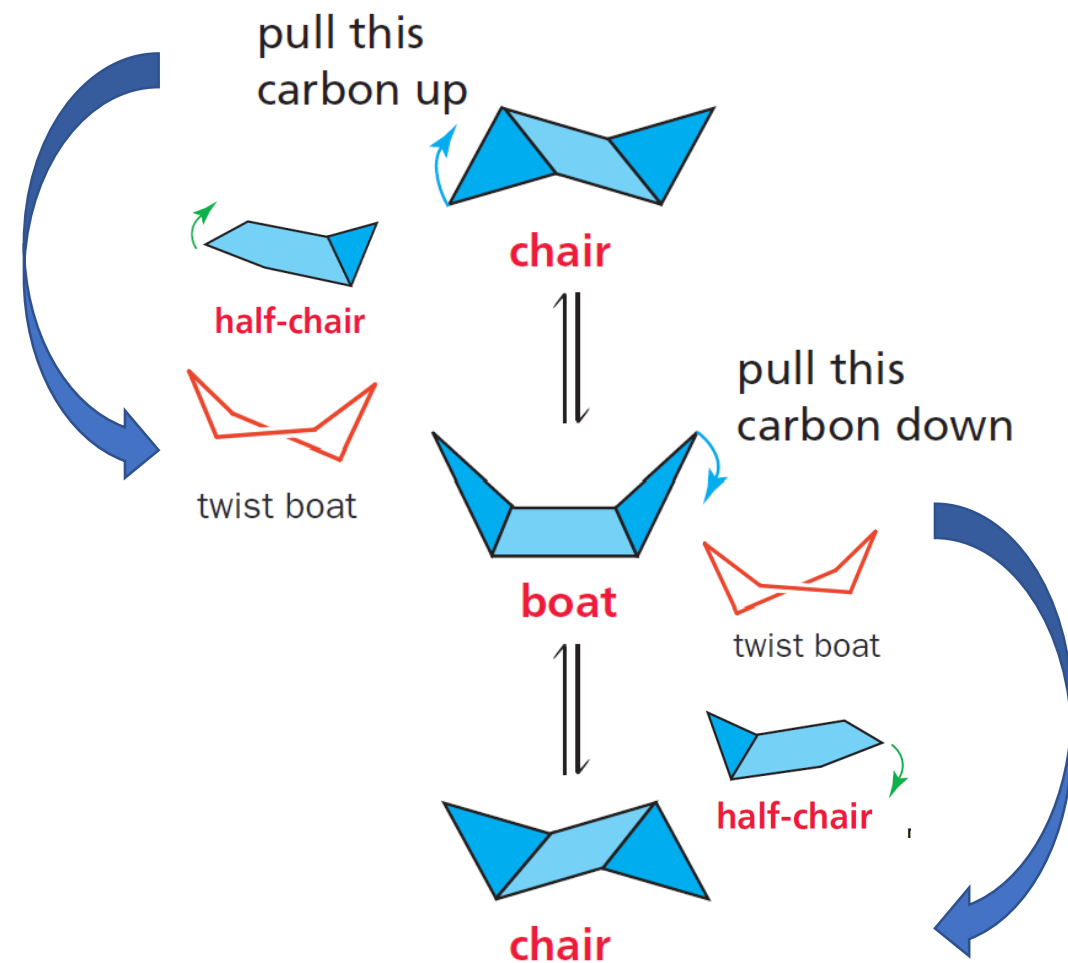
CONFORMERS OF CYCLOHEXANE

- **Cyclohexane** can also exist as a **boat conformer**.
- Like the chair conformer, the **boat conformer** is **free of angle strain**. However, the boat conformer is **not as stable** because some of the **C-H bonds are eclipsed**.
- The boat conformer is further destabilized by the close proximity of the **flagpole hydrogens**—the hydrogens at the “*bow*” and “*stern*” of the boat—which cause **steric strain**.



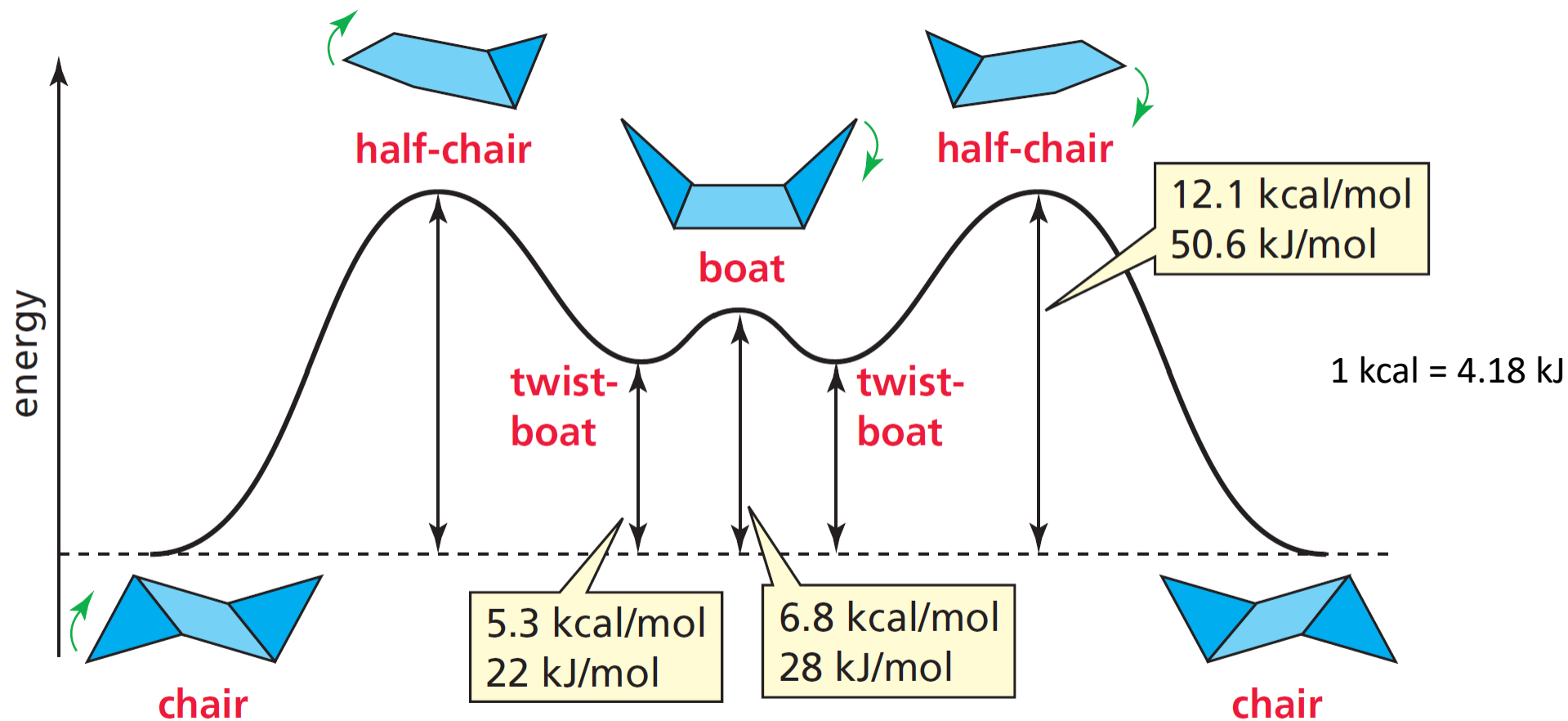
CONFORMERS OF CYCLOHEXANE

- To convert from the **boat conformer** to a **chair conformer**, one of the two topmost carbons of the **boat conformer** must be pulled down so that it becomes the bottom most carbon of the **chair conformer**.
- When the carbon is pulled down just a little, the **twist-boat conformer** is obtained, which is more stable than the boat conformer because the flagpole hydrogens have moved away from each other, thus relieving some steric strain.
- When the carbon is pulled down to the point where it is in the same plane as the sides of the boat, the very unstable **half-chair conformer** is obtained. Pulling the carbon down farther produces the chair conformer.



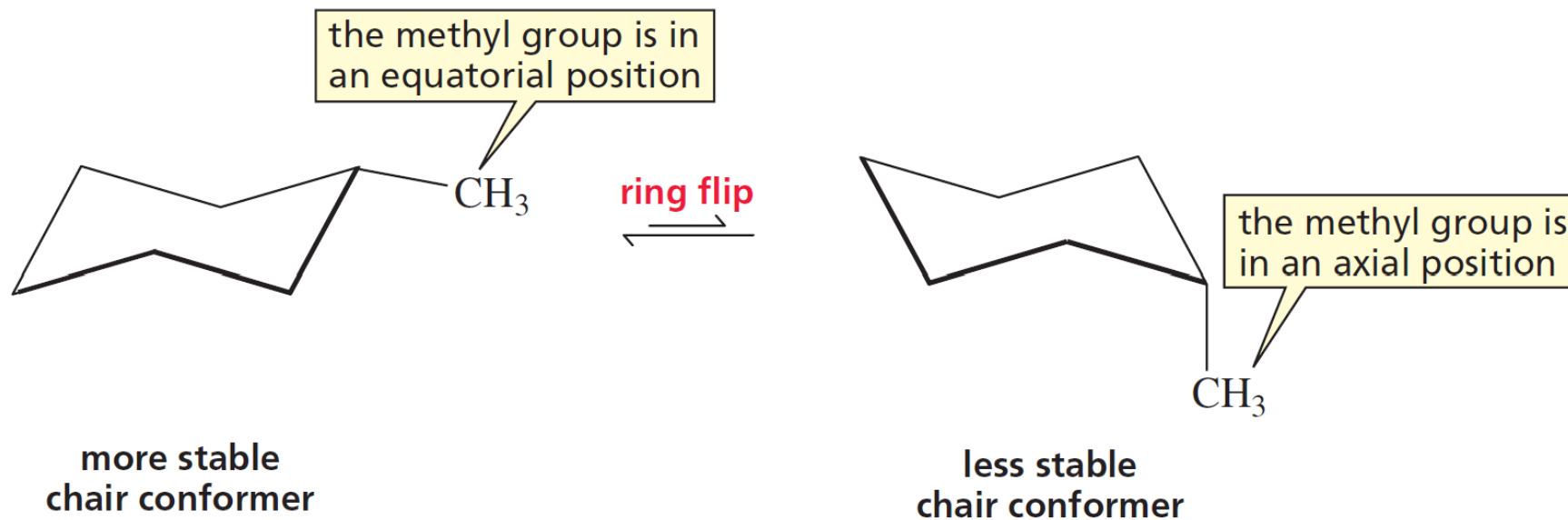
CONFORMERS OF CYCLOHEXANE

- The conformers of cyclohexane—and their relative energies—as one chair conformer interconverts to the other chair conformer.



CONFORMERS OF MONOSUBSTITUTED CYCLOHEXANES

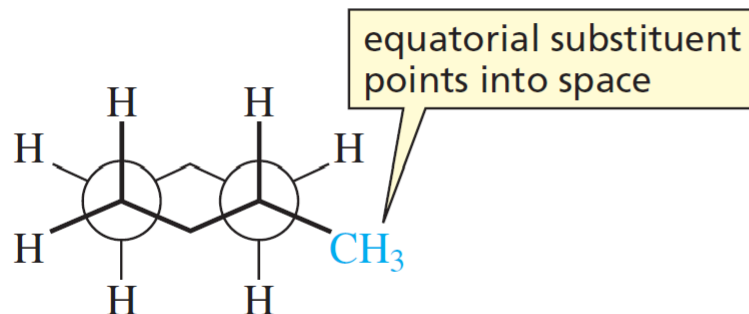
- The two chair conformers of a **monosubstituted cyclohexane** (For e.g. : methylcyclohexane) are not equivalent.
- A substituent is in an **equatorial position** in one chair conformer and in an **axial position** in the other.
- The conformer with the substituent in the **equatorial position** is **more stable**.



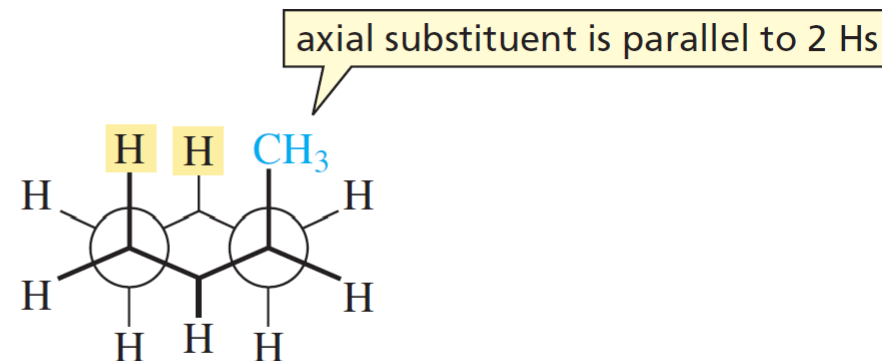
CONFORMERS OF MONOSUBSTITUTED CYCLOHEXANES

The chair conformer with the methyl substituent in an equatorial position is the more stable of the two conformers because a substituent has more room and, therefore, **fewer steric interactions** when it is in an equatorial position.

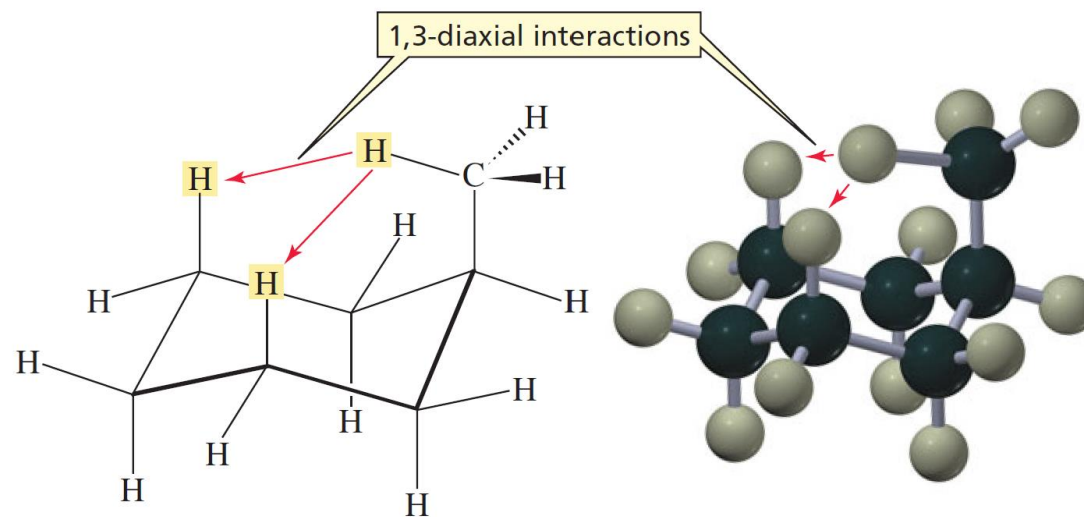
a.



b.

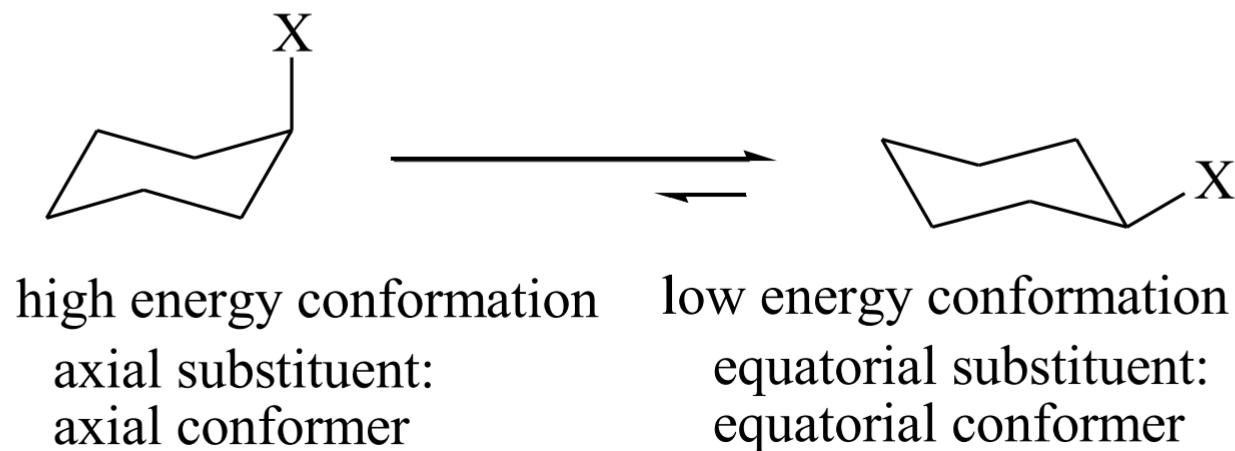


Any **axial** substituent will be relatively close to the axial substituents on the other two carbons on the same side of the ring because all three axial bonds are parallel to each other. Because the interacting axial substituents are in **1,3-positions** relative to each other, these unfavorable steric interactions are called **1,3-diaxial interactions**.



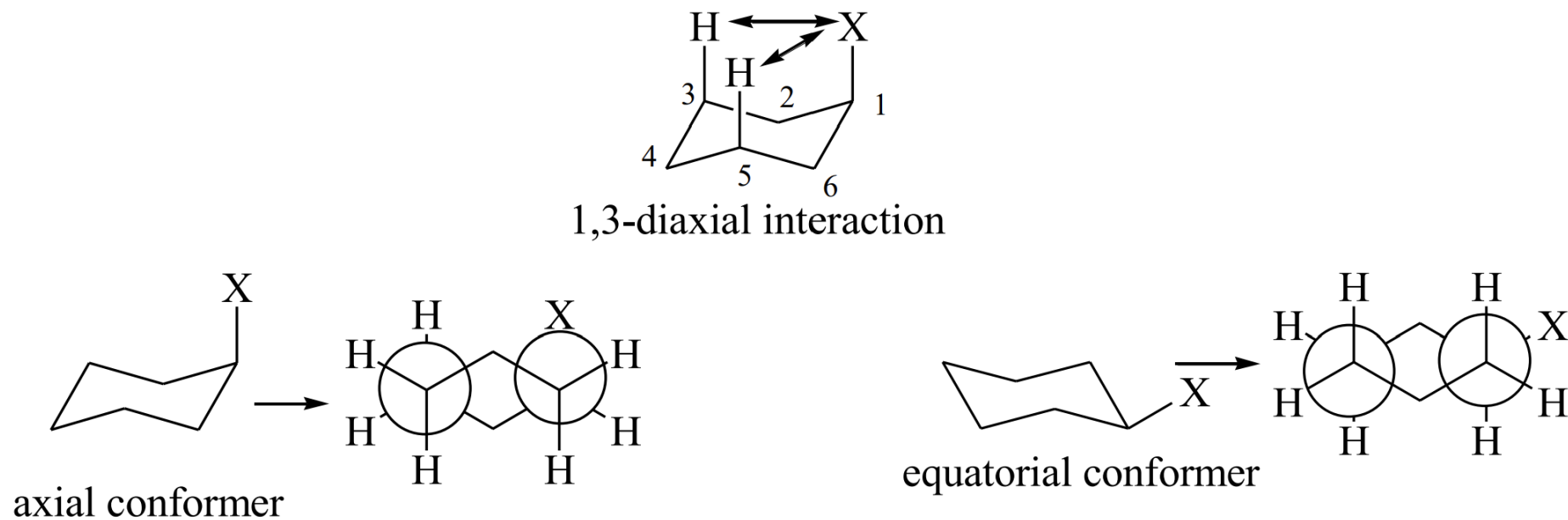
RELATIVE STABILITIES OF CONFORMERS OF MONOSUBSTITUTED CYCLOHEXANES

In case of monosubstituted cyclohexane, two different chair conformer exists-with axial substituents called axial conformer and with equatorial substituents called equatorial conformer. Both conformers are in rapid equilibrium by means of inversion but they will not have same energy. In almost all cases, the conformer with axial substituents has higher energy compared to the equatorial conformer and hence the population percentage of axial conformer is far less than the equatorial conformer.



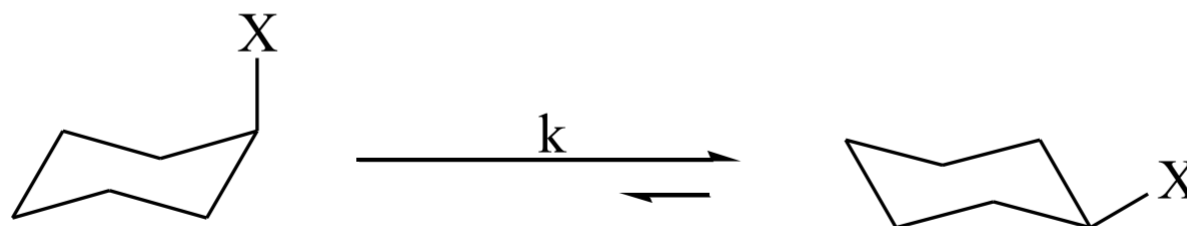
RELATIVE STABILITIES OF CONFORMERS OF MONOSUBSTITUTED CYCLOHEXANES

For $X = -CH_3$, the axial conformer is 7.3 KJ/mol higher energy than the equatorial conformer. At 25°C temperature, axial equatorial ratio is 20:1. This higher energy of axial conformer arises from the 1,3-diaxial interaction between the axial substituents ($-X$) and the axial hydrogens of C3 and C5.



RELATIVE STABILITIES OF CONFORMERS OF MONOSUBSTITUTED CYCLOHEXANES

As the size of X increases, the 1,3-diaxial interaction increases and hence the energy of the axial conformer increases or the percentage population of axial conformer decreases.



RELATIVE STABILITIES OF CONFORMERS OF MONOSUBSTITUTED CYCLOHEXANES

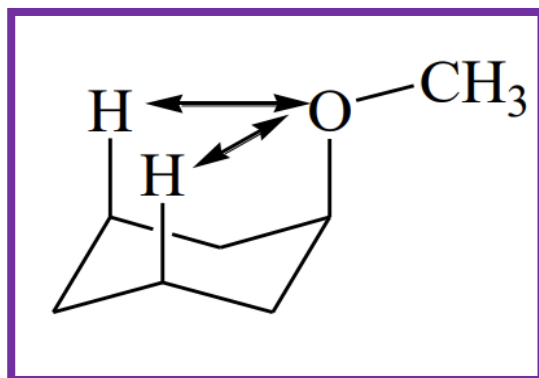
The table below summarizes how the equilibrium constant for axial to equatorial ring inversion depends on different substituent.

Substituents (-X)	Equilibrium constant (k)	KJ/mol (axial-Equatorial)	% of equatorial conformer
-H	1	0	50
-CH ₃	19	7.3	95
-Et	20	7.5	95
- ⁱ Pr	42	9.3	98
- ^t Bu	>3000	>20	>99.9
-OMe	2.7	2.5	73
-Ph	110	11.7	99

From the table it is clear that as the size of the substituents increases, the equilibrium is shifting towards the equatorial conformer.

RELATIVE STABILITIES OF CONFORMERS OF MONOSUBSTITUTED CYCLOHEXANES

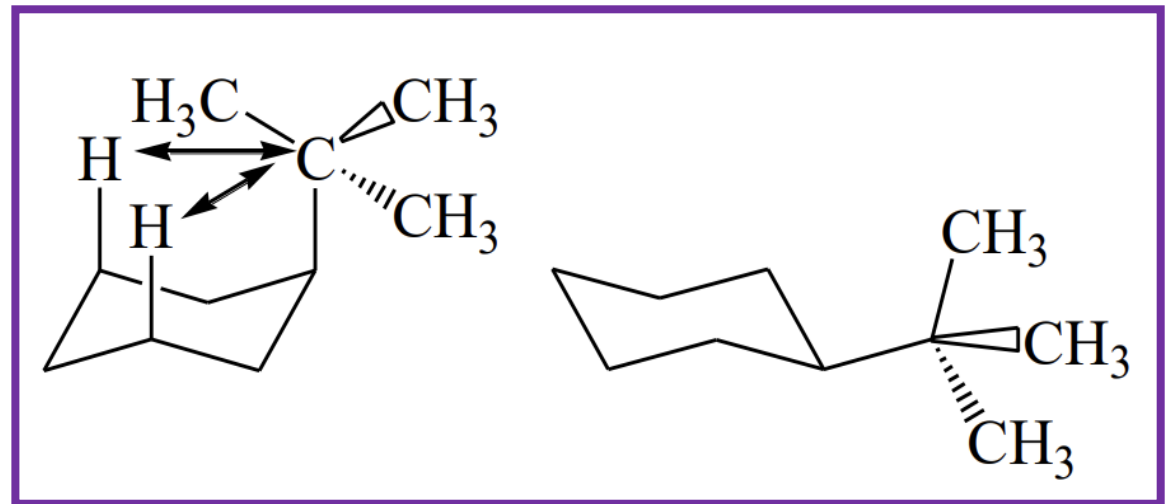
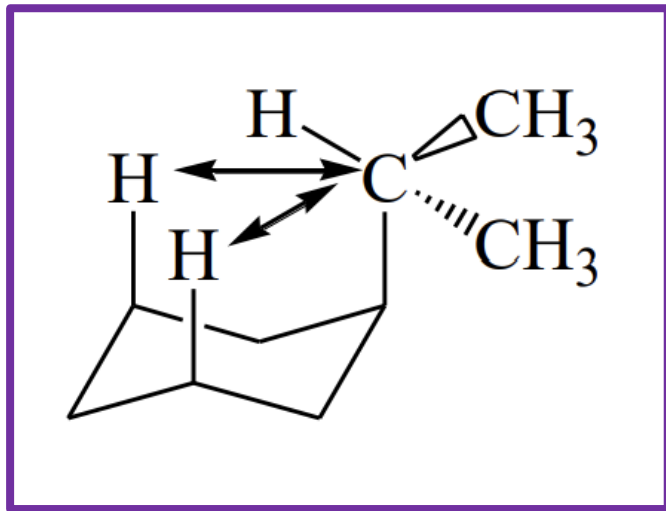
- Although **methoxy group** is **larger** compared to **methyl** or **ethyl** group, still the equilibrium constant for methoxy substituent is substantially less than methyl or ethyl substituted cyclohexane. This is due to the fact that the **1,3 diaxial interaction** is effective between the axial hydrogen and the first atom of the axial substituted group
- In case of methoxy substituent, although the total size of methoxy is larger than methyl, but the **1,3-diaxial interaction** happens between the **axial hydrogen** and the comparatively **smaller oxygen atom** of **methoxy group** and hence the interaction is **less active**.



Substituents (-X)	Equilibrium constant (k)	KJ/mol (axial-Equatorial)	% of equatorial conformer
-H	1	0	50
-CH ₃	19	7.3	95
-Et	20	7.5	95
- ⁱ Pr	42	9.3	98
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-OMe	2.7	2.5	73
-Ph	110	11.7	99

RELATIVE STABILITIES OF CONFORMERS OF MONOSUBSTITUTED CYCLOHEXANES

Unusual high equilibrium constant for **tertiary butyl group** compared to **isopropyl group** can be explained in terms of relative possibility of interaction. In case of **isopropyl group** due to free C-C bond rotation, at **least once the hydrogen interacts with other two diaxial hydrogen** but in case of **tertiary butyl group** every time **the two diaxial hydrogen will meet methyl group** and hence the equilibrium will almost shifted towards the inverted ring where the substituent will be at equatorial position.



RELATIVE STABILITIES OF CONFORMERS OF MONOSUBSTITUTED CYCLOHEXANES

The conformational free energy for **chlorine**, **bromine** and **iodine** is almost same although the size of halogen increases is due to the fact that due to enhancement of size, **the bond length also increases and hence the 1,3-dipolar interaction decreases**.

Conformational free energy for halogen substituents (KCal/mol)

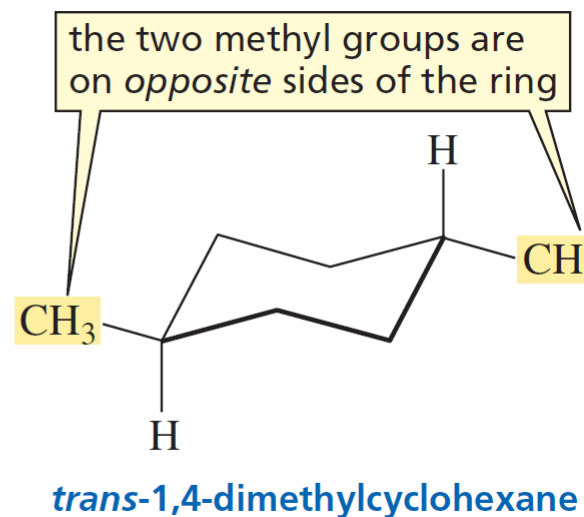
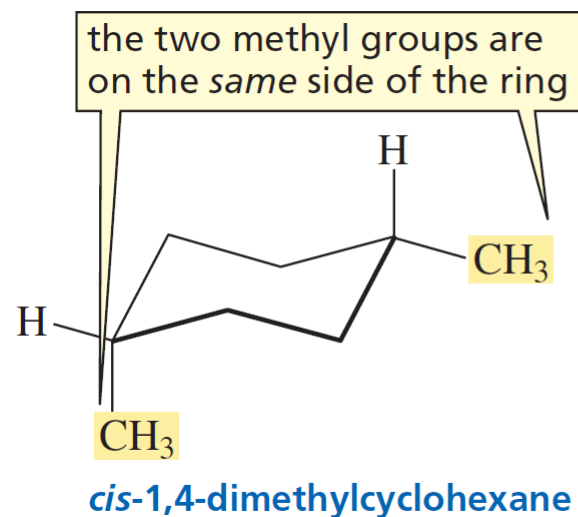
-F	0.25
-Cl	0.53
-Br	0.48
-I	0.51

1 KCal = 4.18 KJ

CONFORMERS OF DISUBSTITUTED CYCLOHEXANES

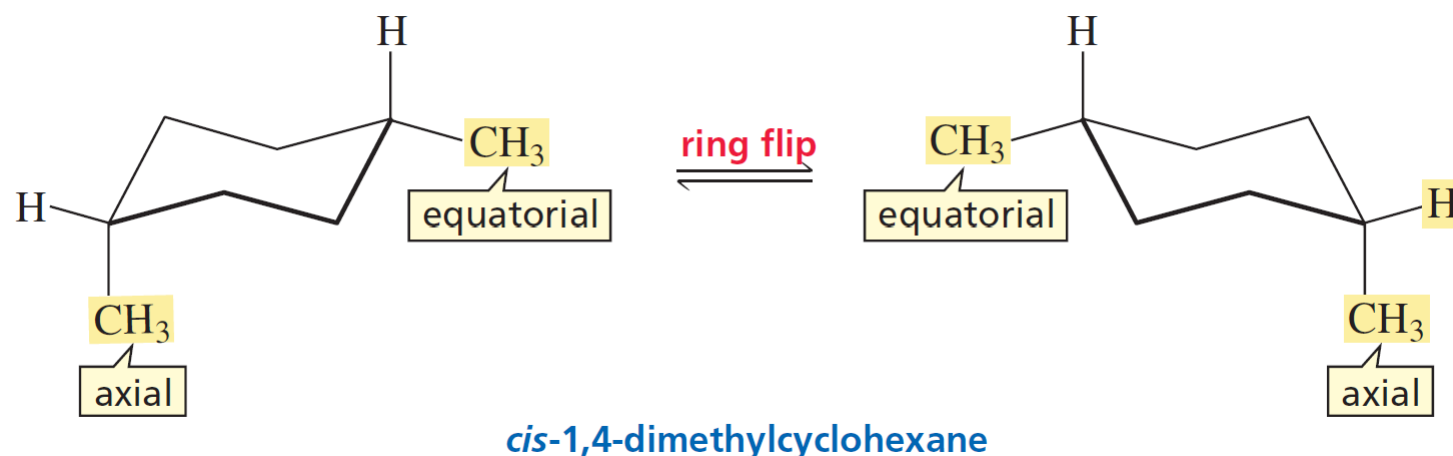
- If a cyclohexane ring has two substituents, we must take both substituents into account when predicting which of the two chair conformers is more stable.
- Let's use **1,4- dimethylcyclohexane** as an example.

First of all, note that there are **two different dimethylcyclohexanes**. One has both methyl substituents on the same side of the cyclohexane ring (both point downward)—it is called the ***cis*** isomer. The other has the two methyl substituents on opposite sides of the ring (one points upward and one points downward)—it is called the ***trans*** isomer.



CONFORMERS OF DISUBSTITUTED CYCLOHEXANES

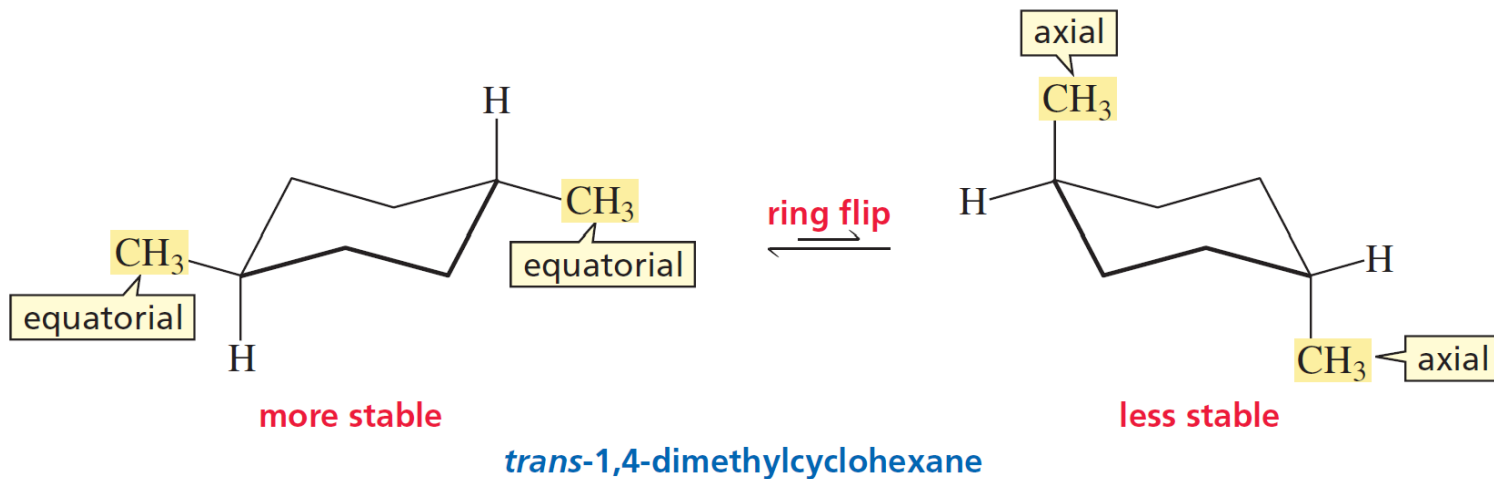
- Every compound with a cyclohexane ring has two chair conformers; thus, both the *cis* isomer and the *trans* isomer of a **disubstituted cyclohexane** have **two chair conformers**.
- Let's compare the structures of the two chair conformers of *cis* -1,4-dimethylcyclohexane to see if we can predict any difference in their stabilities.



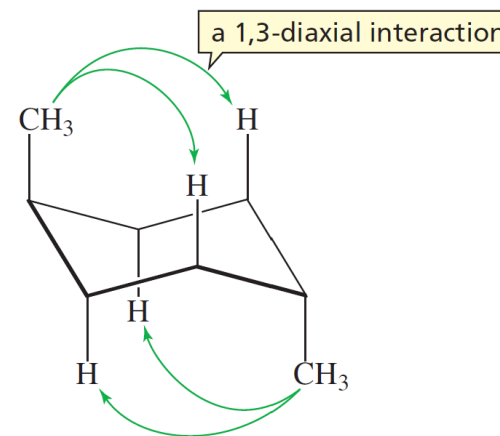
The conformer shown on the left has **one methyl group** in an **equatorial position** and **one methyl group** in an **axial position**. The conformer on the right also has **one methyl group** in an **equatorial position** and one methyl group in an **axial position**. Therefore, both chair conformers are **equally stable**.

CONFORMERS OF DISUBSTITUTED CYCLOHEXANES

In contrast, the two chair conformers of **trans -1,4-dimethylcyclohexane** have different stabilities because one has both methyl substituents in equatorial positions and the other has both methyl groups in axial positions. The conformer with both substituents in equatorial positions is more stable.



The chair conformer with both substituents in **axial positions** has **four 1,3-diaxial interactions**, causing it to be less stable than the chair conformer with both methyl groups in **equatorial positions**. Thus, almost all the molecules of **trans -1,4 dimethylcyclohexane** will be chair conformers with both substituents in equatorial positions.

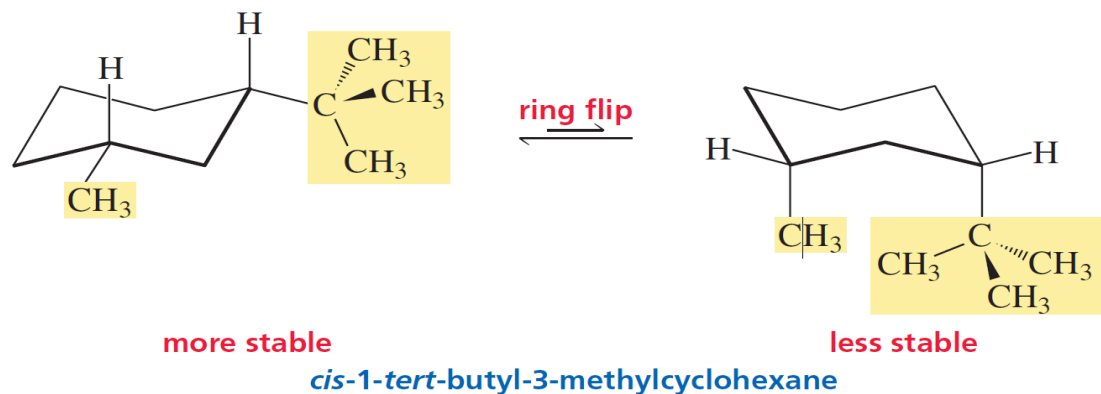


this chair conformer has four 1,3-diaxial interactions

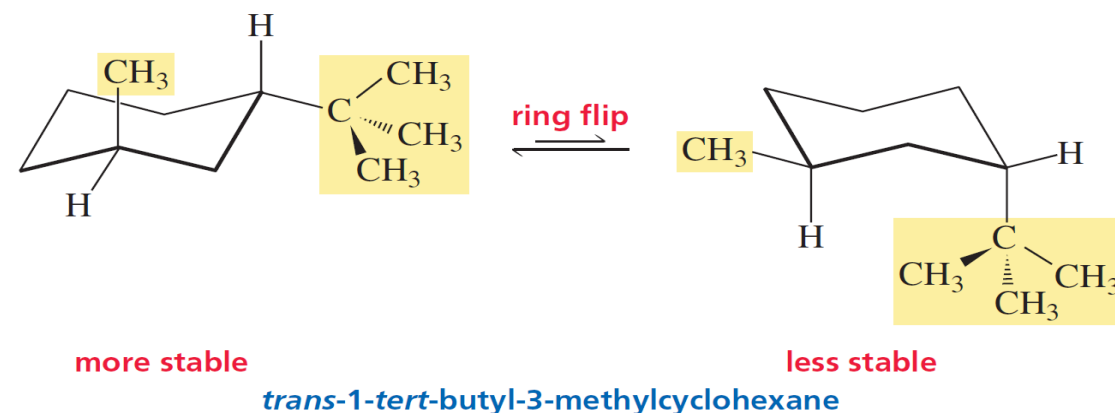
CONFORMERS OF DISUBSTITUTED CYCLOHEXANES

Now let's look at the geometric isomers of **1- *tert* -butyl-3-methylcyclohexane**.

Both substituents of the ***cis* isomer** are in **equatorial** positions in one chair conformer and both are in **axial** positions in the other. The conformer with both substituents in **equatorial positions is more stable**.



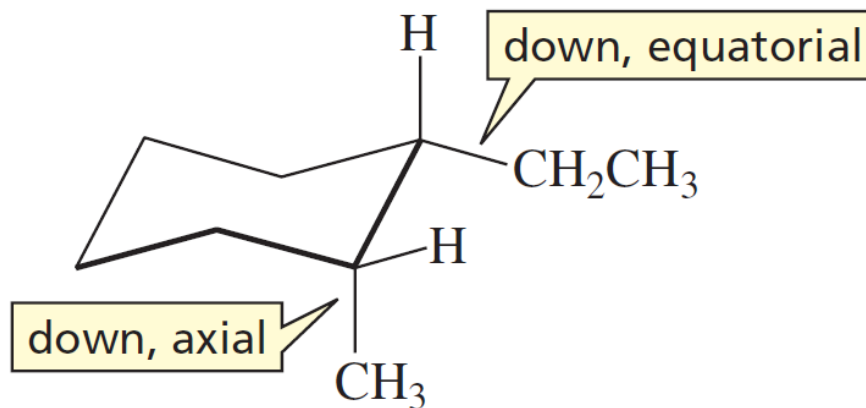
Both chair conformers of the ***trans* isomer** have one substituent in an **equatorial** position and the other in an **axial** position. Because the ***tert* -butyl group** is larger than the methyl group, the **1,3-diaxial interactions** will be stronger when the ***tert* -butyl group** is in an **axial** position. Therefore, the conformer with the ***tert* -butyl group** in an **equatorial position is more stable**.



CONFORMERS OF DISUBSTITUTED CYCLOHEXANES

Draw the more stable chair conformer of *cis* -1-ethyl-2-methylcyclohexane ?

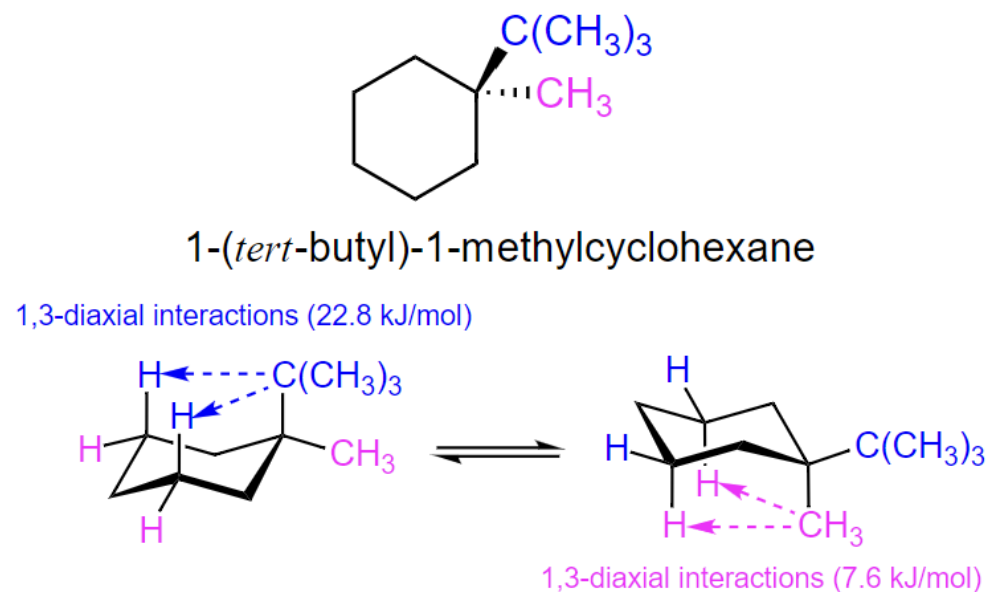
If the two substituents of a 1,2-disubstituted cyclohexane are to be on the same side of the ring, one must be in an equatorial position and the other must be in an axial position. The more stable chair conformer is the one in which the larger of the two substituents (the ethyl group) is in the equatorial position.



CONFORMERS OF DISUBSTITUTED CYCLOHEXANES

1,1-Disubstituted Cyclohexanes

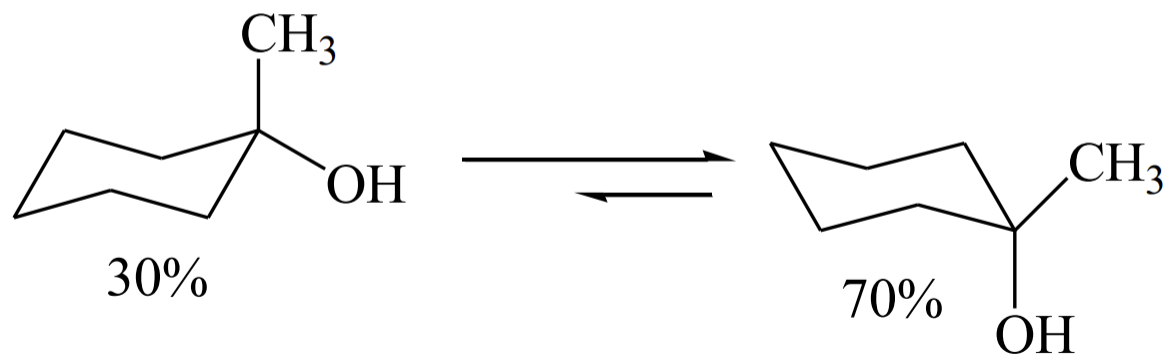
- If the two groups are different, as in **1-*tert*-butyl-1-methylcyclohexane**, then the equilibrium favors the conformer in which the **larger group (*tert*-butyl in this case)** is in the **more stable equatorial position**.



CONFORMERS OF DISUBSTITUTED CYCLOHEXANES

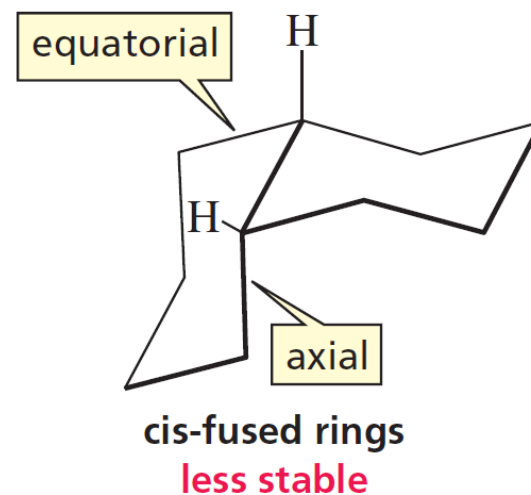
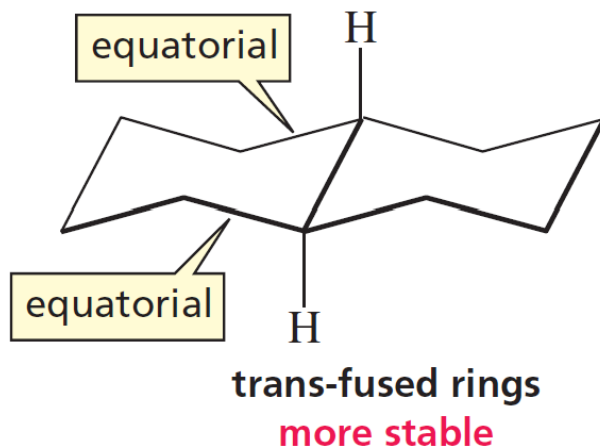
Conformation of 1-methyl cyclohexanol

The conformational free energy of -OH group (0.60 Kcal/mol) is less than -CH_3 (1.73 Kcal/mole); hence -CH_3 group at equatorial position will be preferred over equatorial -OH group.



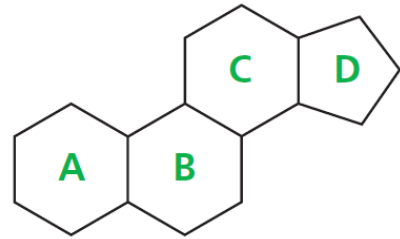
FUSED CYCLOHEXANE RINGS

- When two cyclohexane rings are fused— fused rings share two adjacent carbons—one ring can be considered to be a pair of substituents bonded to the other ring.
- As with any disubstituted cyclohexane, the two substituents can be either ***cis*** or ***trans***.
- The ***trans* isomer** (in which one substituent bond points upward and the other downward) has both substituents in the **equatorial position**.
- The ***cis* isomer** has one substituent in the equatorial position and one in the axial position.
- ***Trans***-fused rings, therefore, are **more stable** than ***cis***-fused rings.

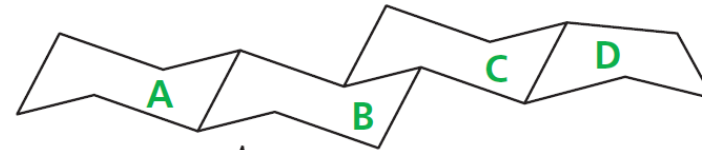


FUSED CYCLOHEXANE RINGS

- **Hormones** are chemical messengers—organic compounds synthesized in glands and delivered by the bloodstream to target tissues in order to stimulate or inhibit some process. Many hormones are **steroids**.
- **Steroids** have four rings designated here by **A, B, C, and D**. The **B, C, and D** rings are all **trans fused**, and in most naturally occurring steroids, the **A and B** rings are also **trans fused**.



the steroid ring system



all the rings are trans fused

- The most abundant member of the steroid family in animals is **cholesterol**, the precursor of all other **steroids**.
- **Cholesterol** is an important component of **cell membranes**. Because its rings are locked in a specific conformation, it is more rigid than other membrane components.

